

TEMASEK POLYTECHNIC
SCHOOL OF INFORMATICS & IT
DIPLOMA IN CYBER & DIGITAL FORENSICS
AY2023/2024 OCTOBER SEMESTER (LEVEL 2) TERM A
CLOUD SECURITY (CCF2C05)
Project (70%)

CLOUD SECURITY (CCF2C05)

SUBJECT LEVEL: 2

INSTRUCTIONS TO CANDIDATES

1. This project is a group project.
2. Ideally form 6 students per group.
The project should be equally divided by the group members.
3. Project (70%).

No.	Component	Assess by	Weightage %	Complete by
1	Certificate of Cloud Security Knowledge (CCSK)			
a	Group Presentation Slide	Group	10	Week 7
b	Presentation	Individual	10	Week 7
2	IoT Project			
a	Project Final Report	Group	10	Week 9
b	Project Presentation, Demo and Q&A	Individual	30	Week 9
c	AWS Cloud Foundations Certificate	Individual	10	Week 9
	Total		70	



Scenario and Scope

Part 1. Certificate of Cloud Security Knowledge (CCSK)

You are a new hire IoT Cloud Engineer, before embarking on the IoT Cloud project, you are required to demonstrate your Cloud Security knowledge. Your group will be required to present to the class on your selected domains found in the Certificate of Cloud Security Knowledge (CCSK) [Security Guidance v4](#) reference material. The domains to be presented will be divided among the class grouping. The chosen domains should be unique for each group and ideally not repeated by other class groups.

TABLE OF CONTENTS			
DOMAIN 1 Cloud Computing Concepts and Architectures	DOMAIN 2 Governance and Enterprise Risk Management	DOMAIN 3 Legal Issues, Contracts and Electronic Discovery	DOMAIN 4 Compliance and Audit Management
DOMAIN 5 Information Governance	DOMAIN 6 Management Plane and Business Continuity	DOMAIN 7 Infrastructure Security	DOMAIN 8 Virtualization and Containers
DOMAIN 9 Incident Response	DOMAIN 10 Application Security	DOMAIN 11 Data Security and Encryption	DOMAIN 12 Identity, Authentication, and Access Management
DOMAIN 13 Security as a Service	DOMAIN 14 Related Technologies		

CCSK Power Point Presentation (Group 10% + Individual 10% = 20%)

- Select the domains to present
- Develop concise and easy to follow study notes that can be used for preparing the CCSK certification
- Well formatted, readable, and aesthetically good
- ~30 min per group

Part 2. IoT PROJECT

You are to complete all the AWS Knowledge Checks and Assignments given in the respective AWS modules. You are required to be **certified in the AWS Cloud Foundation** module to demonstrate your Cloud Foundation knowledge while embarking on the IoT Project.

IoT systems often consist not only of sensor and actuator electronics, but also of software applications that control the systems through a cloud platform. In home or industrial automation systems, sensors and actuators can be connected to the local network or over Wi-Fi to a cloud management platform through the Internet. The cloud management platform collects data from the IoT system and enables control of it from either a browser or specialized app that runs on a mobile device.

Many IoT systems include web applications that are used to monitor and control IoT devices. Many of these applications are accessed across the open Internet. Users authenticate to the application to make use of it. IoT devices use web applications to provide users with an effortless way to monitor sensors and control actuators such as thermostats, valve, PLC, electrical outlets, and alarms.

Message queuing telemetry transport (MQTT) is an extremely simple and lightweight publish/subscribe messaging protocol. It is designed for IoT devices and uses minimal bandwidth and device resources. MQTT is an ideal protocol for machine-to-machine (M2M) communications. MQTT uses the reserved TCP/IP port 1883 by default. When using MQTT over SSL, it uses the registered TCP/IP port 8883.

Your group is to implement a proof of concept (PoC) example of an IoT home automation using one of the following implementations as depicted below:

IoT home automation and monitoring system using the AWS Cloud platform

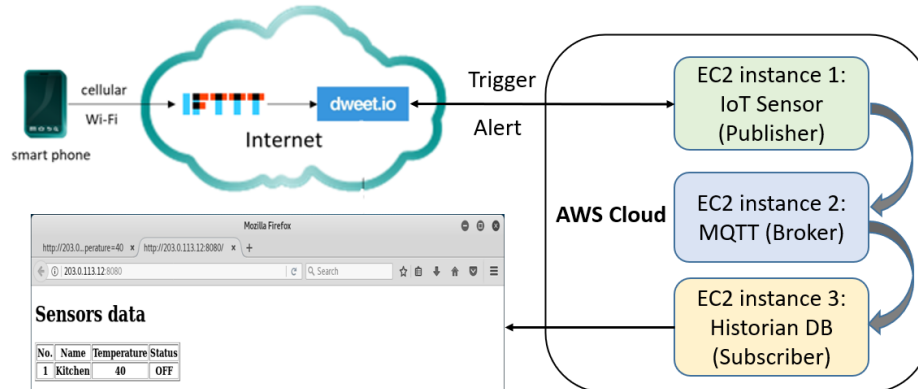


Figure 1. IoT system using AWS Cloud VM

The Historian DB should have the sensor data displayed such as the following:

No.	Time Stamp	Device	Status	Reading
1	2020.03.23-10:00:00	Air-con	On	Temp=25
2	2020.03.23-13:00:00	Light	Off	Brightness=0

In this project, you are to define and simulate an IoT system as depicted in figure 1. Example, a home automation appliances temperature control and status monitoring system.

You should be able to trigger sensor events or receive alerts via a mobile device through the integration of sensor(s), controller (e.g., **emulating using the virtual Pi**), AWS cloud and web services (e.g., IFTTT). The sensors readings will be taken and stored into a Historian DB server for display in HTML. MQTT protocol will be used for communication between the sensors and DB.

To do well and distinct yourself, you are strongly encouraged to go beyond the course materials and explore other relevant AWS cloud services e.g., Fargate, IoT services or tools that can be used in the context of your project.

When evaluating system security, it is useful to break the system down into parts in order to identify different areas of potential vulnerability. You will analyse the system to identify potential vulnerabilities/risks at the device, communication, and application layers of the simplified IoT System and implement/recommend security measures to mitigate them.

Note:

This project is an integration of the IoT lab exercises you have done. You may need to tweak and integrate the lab exercises using third party tools like paho-mqtt client to complete the project. The instructor may provide little or no guidance on the technical issues encountered. This is to assess your technical resolution skill.



Deliverables

2.1 PROJECT REPORT

Please reference report template given.

NOTE:

Backup your works for the demo assessment. Any reason such as corrupted OS, unable to boot etc., that hinder the conduct of the demo assessment will result in ZERO mark for the components being assess.

2.2 EXPECTATIONS

IoT Project Report, Demo and Q&A (50%) able to:

- a) Group report on the IoT System and its security considerations.
- b) Present the proposed IoT solution.
- c) Demonstrate the integrated IoT System on a cloud platform.
- d) Articulate a good understanding of Cloud and IoT Security concepts.
- e) Attain AWS modules certificate of completion.

Plagiarism

Plagiarism is not allowed and will result in a zero for the assignment; **including AI generated work e.g., ChatGPT**. Please cite any sources used in your research. It is expected that each group member contributes equally.

The LMS will highlight similar sentences that are copied from the websites and reports submitted by your classmate as plagiarism. Do note that we treat plagiarism seriously; hence disciplinary actions (E.g., failing the subject) will be taken.

Penalties for late submissions

late and < 1 day: 10% deduction from absolute mark given for the assignment e.g., 75 marks (100 marks max) becomes 65 marks (deduct 10% of 100 marks)

late ≥ 1 and < 2 days: 20% deduction from absolute mark

late ≥ 2 days: 0 marks awarded

-END-